Category Theory for Mathematical Physics — Exercise Sheet 3

Monoidal categories and string diagrams

Variant: Questions only

Conventions as in Lecture 3: pictures read upward, the first factor on the left; Vec_G^ω and its cocycle condition as in Definition 3.7 and Theorem 3.8. The group $\mathbb{Z}_N$ is written additively, with elements $0, 1, \dots, N-1$ and $[x]_N$ the remainder of an integer x mod N; $\zeta_N = e^{2\pi i/N}$. In Exercise 4, as in Lecture 3, $\mathbb{Z}_2 = \{1, s\}$ is written multiplicatively, and the *semion category* is Vec_G^ω on it with $\omega(s,s,s) = -1$. Here ε is the carry of Exercise 3 (not the toric-code anyon); e and u are the units of two operations in Exercise 2, and u is a unit of $\mathbb{Z}_N$ in Exercise 6; in Exercise 5, φ is a group automorphism (not the golden ratio).

Exercise 1 (Interchange and pictures). Let $f\colon X \to X'$, $g\colon X' \to X''$, $h\colon Y \to Y'$ and $k\colon Y' \to Y''$.

(i) Draw $(g \otimes k) \circ (f \otimes h)$ and $(g \otimes \mathrm{id}_{Y''}) \circ (\mathrm{id}_{X'} \otimes k) \circ (f \otimes h)$, and show that both equal $(g \circ f) \otimes (k \circ h)$.

(ii) In $\mathrm{Mat}_{\mathbb{C}}$ let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Compute $A \otimes B$ and $B \otimes A$ (Kronecker products), and check that $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$.

(iii) Boxes on different strands slide past each other (Equation (3.3)). Can two boxes on the same strand? Answer with A and B.

Exercise 2 (Eckmann–Hilton, in full). Let M be a set with two binary operations $\cdot$ and $*$, with units u for $\cdot$ and e for $*$, such that

$$(a * b) \cdot (c * d) = (a \cdot c) * (b \cdot d) \quad \text{for all } a, b, c, d \in M.$$

(i) Show that $u = e$.

(ii) Show that $a \cdot b = a * b$ and $a \cdot b = b \cdot a$ for all a, b.

(iii) Apply this to $\mathrm{End}(1)$ in a strict monoidal category, with $\cdot = \circ$ and $* = \otimes$, to recover Theorem 3.6, and draw the computation of (ii) for $s, t \in \mathrm{End}(1)$ as pictures of boxes with no strands. Where does strictness enter?

Exercise 3 (The cocycles ω_p of $\mathbb{Z}_N$). For $b, c \in \{0, \dots, N-1\}$ let $\varepsilon(b,c) = (b + c - [b+c]_N)/N$, the *carry* of $b + c$, and let $\omega_p(a,b,c) = e^{2\pi i\, p\, a (b+c-[b+c]_N)/N^2}$.

(i) Show that $\varepsilon(b,c) \in \{0,1\}$ and $\omega_p(a,b,c) = \zeta_N^{\,p\,a\,\varepsilon(b,c)}$. Hence ω_p depends only on p mod N.

(ii) Show that $\varepsilon(b,c) + \varepsilon(a, [b+c]_N) = \varepsilon(a,b) + \varepsilon([a+b]_N, c)$.

(iii) Draw the pentagon of Definition 3.1 for simple objects g, h, k, m of Vec_G^ω, with each vertex labelled by its bracketing (each is the object δ_{ghkm}) and each arrow by its scalar, and the triangle for $g, 1, h$ likewise; read off the conditions of Theorem 3.8. Then show that ω_p is normalised and satisfies the cocycle condition, which for $\mathbb{Z}_N$ written additively reads

$$\begin{aligned} &\omega(a+b,c,d)\,\omega(a,b,c+d) \\ &\quad = \omega(a,b,c)\,\omega(a,b+c,d)\,\omega(b,c,d). \end{aligned}$$

(iv) For $N = 3$ and $p = 1$, list the triples (a,b,c) with $\omega_1(a,b,c) \neq 1$, and the values there.

Exercise 4 (The forgetful functor as a monoidal functor). (i) Show that the forgetful functor $U\colon \mathrm{Rep}(G) \to \mathrm{Vect}_{\mathrm{fd}}$, with J and J_0 identities, is monoidal (Definition 3.10).

(ii) Let $F\colon \mathrm{Vec}_G^{\omega} \to \mathrm{Vect}_{\mathrm{fd}}$ send a graded space to its underlying space, so $F(\delta_g) = \mathbb{C}$, and give it $J_{g,h} = j(g,h)\,\mathrm{id}_{\mathbb{C}}$ (with $\mathbb{C}\otimes\mathbb{C} = \mathbb{C}$) and $J_0 = j_0\,\mathrm{id}_{\mathbb{C}}$, for nonzero numbers $j(g,h)$, j_0. Show that (F, J, J_0) is monoidal exactly when $\omega = \delta j$, with δj as in Lecture 3, and $j(1,g) = j(g,1) = j_0^{-1}$.

(iii) Show that for $G = \mathbb{Z}_2$ and $\omega(s,s,s) = -1$ no such (F, J, J_0) exists.

Exercise 5 (Monoidal equivalences of Vec_G^{ω}). Let ω and ω' be normalised 3-cocycles on a finite group G. You may use that an equivalence is full, faithful and essentially surjective (Leinster Prop. 1.3.18, the converse of Proposition 1.17).

(i) Let φ be an automorphism of G and $j\colon G\times G \to \mathbb{C}^{\times}$ satisfy

$$\begin{aligned} &\omega'(\varphi(g),\varphi(h),\varphi(k)) \\ &\quad = \omega(g,h,k)\,\frac{j(gh,k)\,j(g,h)}{j(h,k)\,j(g,hk)} \end{aligned} \tag{$*$}$$

for all g,h,k. Show that $F(\delta_g) = \delta_{\varphi(g)}$, extended to direct sums, with $J_{g,h} = j(g,h)$ and $J_0 = j(1,1)^{-1}$, is a monoidal equivalence $\mathrm{Vec}_G^{\omega} \to \mathrm{Vec}_G^{\omega'}$.

(ii) Conversely, let (F, J, J_0) be a monoidal equivalence $\mathrm{Vec}_G^{\omega} \to \mathrm{Vec}_G^{\omega'}$. Show that there are an automorphism φ and a function j satisfying $(*)$.

(iii) Conclude the classification stated in Lecture 3: Vec_G^{ω} and $\mathrm{Vec}_G^{\omega'}$ are monoidally equivalent exactly when ω' is obtained from ω by an automorphism of G and a coboundary.

Exercise 6 ⋆. *The classes of $\mathrm{Vec}_{\mathbb{Z}_N}^{\omega}$ up to equivalence.* For a normalised 3-cocycle ω on $\mathbb{Z}_N$ and $a \in \mathbb{Z}_N$ let

$$Q_{\omega}(a) = \prod_{r=0}^{N-1} \omega(a, ra, a),$$

with ra reduced mod N. Two cocycles are *cohomologous* if $\omega' = \omega\,(\delta j)^{-1}$ for some $j\colon \mathbb{Z}_N^2 \to \mathbb{C}^{\times}$ (Lecture 3).

(i) Show that $Q_{\omega'} = Q_{\omega}$ when ω' and ω are cohomologous.

(ii) Show that $Q_{\omega_p}(1) = \zeta_N^{\,p}$. Conclude that $\omega_0, \dots, \omega_{N-1}$ are pairwise not cohomologous.

(iii) For a unit u mod N let $(u^*\omega)(a,b,c) = \omega(ua, ub, uc)$, the cocycle moved by the automorphism $a \mapsto ua$. Show that $Q_{u^*\omega}(1) = Q_{\omega}(u)$ and $Q_{\omega_p}(u) = \zeta_N^{\,pu^2}$, with u taken in $\{1, \dots, N-1\}$.

(iv) You may use that every normalised 3-cocycle on $\mathbb{Z}_N$ is cohomologous to some ω_p (EGNO Ex. 2.6.4). With Exercise 5, show that $\mathrm{Vec}_{\mathbb{Z}_N}^{\omega_p}$ and $\mathrm{Vec}_{\mathbb{Z}_N}^{\omega_{p'}}$ are monoidally equivalent exactly when $p' \equiv pu^2 \bmod N$ for some unit u, and count the classes for $N = 2, 3, 4, 5$.